Smart Surveillance for Millet Health: An Innovative System for Disease Detection

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Abstract— Millet crops are vital in sustainable agriculture and food security, particularly in semi-arid areas. Their production is highly impacted by fungal and bacterial diseases that would often go unnoticed at early stages because farmers lack accessible diagnostic systems to detect them. This work proposes a hybrid deep learning architecture that combines MobileNetV2 for disease classification and another stage detection model to decide treatability status (treatable or non-treatable). Both models were trained and tested on a handpicked dataset of 14 millet disease classes with high classification accuracy and fast performance on mobile devices. The learned models were incorporated into an API in the form of Flask and offered through an Android app, allowing real-time disease detection based on images directly from smartphone cameras or uploads from the gallery. The system also offers automated treatment advice, farmer-friendly chatbot interface. Experimental evaluations prove that the system can provide consistent predictions within seconds, presenting an efficient, low-cost, and scalable digital tool for rural farming societies. The suggested method bridges the gap between research in deep learning and actual deployment in agriculture, arming farmers with practical knowledge for sustainable millet crop management.

Keywords — Millet disease detection, Deep learning, MobileNetV2, Stage classification, Android application, Computer vision, Smart agriculture, Machine learning, Farmer support system, AI in agriculture.

I. INTRODUCTION

Millets — a generic name for a cluster of hardy, small-grained cereal crops — are among the most climate-tolerant crops of the world. They are particularly well-adapted to semi-arid and marginal habitats, where fewer cereals can survive. Appreciation of their nutritional and environmental value has accelerated: thus, between 2019 and 2021, global millet production rose from about 28.33 million metric tons to about 30.08 million metric tons, according to the Food and Agriculture Organization (FAO). India is still the world's largest producer, accounting for over 40 % of the world's millet production.

Even with such significance, millet production and area under cultivation are still much short of their potential. As per one report, the area under millets in India fell from an approximate 34.8 million hectares during the 1950s to approximately 13.6 million hectares in 2020 — a decline of close to 60 %.

The decline in area and low productivity pose/a twin challenge: (i) food-nutritional security among rural and marginal communities, and (ii) improving farmer livelihoods in rain-fed, dry-land agriculture systems. In the process, management of diseases becomes a serious bottleneck. Millet crops are prey to several fungal, bacterial and viral diseases — including downy mildew, rusts, smuts and leaf blights — which tend to go undetected until the late stages, particularly in resource-poor farming systems. Early disease detection, combined with timely intervention, is therefore critical.

Conventional disease surveillance and advisory processes tend to include agronomists or extension agents making field observations — an activity that is time-consuming, expensive and hard to scale, especially in areas that are far-flung and rural where infrastructure is weak. In contrast, the widespread expansion of smartphones even in rural areas offers a very different prospect. Building on-board deep learning models for image-based disease diagnosis can narrow the divide between research in the laboratory and application in the field.

To bridge this gap, the present paper suggests a hybrid deep-learning-based framework for millet disease identification and support to farmers. The method incorporates a first-level classification model (using MobileNetV2) for detection of disease categories and, as a second level, treatability status (treatable or non-treatable). These models are incorporated into an Android application operating offline (device-based) that is supplemented by a web interface for large-scale farmer reach. In addition, the system generates actionable treatment advice and a natural language chatbot interface, thus translating diagnostic output into farmer-relevant action.

II. LITRATURE REVIEW

The area of plant disease diagnosis has witnessed extreme growth following the development of deep learning and precision agriculture technologies. Recent studies have proved the capabilities of machine vision, Internet of Things (IoT), and mobile computing in turning traditional crop monitoring into data-driven, smart systems.

Mishra et al. [1] suggested a smart and sustainable millet crop monitoring framework with a focus on predictive intelligence for early disease identification. Their solution integrated edge computing with cloud-supported analytics to inform real-time disease management. Likewise, Kundu et al. [2] built an IoT-based and interpretable machine-learning framework for pearl millet disease prediction, demonstrating how to combine sensor data and environmental features to enhance classification performance and model decision transparency.

Chaturvedi et al. [3] investigated classical and deep learning approaches for the identification of disease from pearl millet leaf images and compared models like CNN and SVM for multi-class classification. Their results confirmed the reliability of deep convolutional architectures compared to traditional image processing techniques. Pramitha et al. [4] further developed this idea by integrating deep learning models into mobile devices, proving the implementability of pearl millet disease diagnosis using smartphone apps for farmers.

At a larger level, Waldamichael et al. [5] gave an overview of machine learning in the detection of cereal crop diseases and identified issues like inadequate diversity in datasets, poor model generalization, and model interpretability. Shahi et al. [6] discussed recent developments with the use of unmanned aerial vehicles (UAVs) coupled with deep neural networks for monitoring on a large field scale. They suggested UAV imagery, when subjected to processing by CNNs, could provide scalable solutions for detecting disease under heterogeneous conditions of fields.

Li et al. [7] provided an overall review of deep learning plant disease detection methods and concluded that transfer learning models such as ResNet and MobileNet dramatically minimize the requirement of huge labeled datasets. Concomitantly, Zanzaney et al. [8] compared different deep neural network architectures for generalizable crop disease classification and revalidated the benefit of convolutional layers in extracting intricate spatial features in leaf textures.

The initial work by Mohameth et al. [9] using the PlantVillage dataset reaffirmed feature extraction and data augmentation as crucial methods to enhance model accuracy for several species of plants. Kulkarni et al. [11] and Ahmed & Yadav [12] extended this further by using image processing and machine learning algorithms on different crops, ascertaining effective disease identification under a variety of lighting and background conditions.

Ahmad et al. [13] offered a comprehensive overview of deep learning technology in plant disease diagnosis and suggested pragmatic guidelines for developing field-deployable, scalable applications, including data acquisition, preprocessing, and device consideration. In parallel, Morbekar et al. [14] showed the effectiveness of YOLO (You Only Look Once) models for real-time localizing crop disease and how one-shot detectors can trade off between speed and accuracy in field settings.

The collective evidence from such investigations indicates a persistent direction toward hybrid systems fusing computer vision, IoT, and mobile technology for management of plant diseases in agriculture. Yet, the majority of the available work either deals with laboratory experiments only or demands sustained internet connectivity for inference, restricting their feasibility in rural areas with poor connectivity.

To fill these gaps, this work proposes a hybrid deep learning system specifically for millet crops. It combines a two-model pipeline of one for disease classification and another for prediction of treatability within an offline executable Android application. In contrast to previous works, the system not just offers disease diagnosis but also prescriptive treatment advice, voice-controlled chatbot interaction, and web-based integration for large-scale outreach to farmers.

III. PROPOSED METHODOLOGY

The system to be proposed uses a hybrid deep learning architecture that is specifically intended to identify and categorize millet leaf diseases precisely, as well as to evaluate their treatability status. The design combines a dual-model prediction pipeline, farmer-support chatbot, and an Android-based mobile application for real-time, offline disease detection and solution provision. The whole approach can be delineated into five primary steps: data acquisition and preprocessing, model training, dual-model integration, system deployment, and user interaction via the mobile and web interface.

a. Data Collection and Dataset Composition

An exhaustive dataset was compiled comprising high-quality millet leaf images gathered from farm research archives, open source repositories, and on-ground acquisitions in proximity to riverbanks and farms. Three varieties of millet — Finger (Ragi), Pearl, and Sorghum (Jowar) — were covered in the dataset.

Every image was labeled and grouped into 14 disease classes, as detailed below:

Finger (Ragi) Downy, Finger (Ragi) Mottle, Finger (Ragi) Smut, Finger (Ragi) Wilt, Finger (Ragi) Seedling, Finger (Ragi) Healthy, Pearl Healthy, Pearl Rust Disease, Pearl Downy Mildew, Sorghum (Jowar) Blast, Sorghum (Jowar) Ergot, Sorghum (Jowar) Rust, Sorghum (Jowar) Smut, Sorghum (Jowar) Healthy.

The data was then restructured again into a modified binary dataset, which was referred to as Dataset_Modified, containing two superclasses — Treatable and Non-Treatable. The treatable class had early-stage or manageable infections (e.g., Rust, Downy, Mildew), whereas the non-treatable class had irreversible or late-stage diseases (e.g., Wilt, Smut). This hierarchical labeling allowed the model to predict both the disease name and its treatability.

b. Data Preprocessing

All acquired images were normalized using a comprehensive preprocessing pipeline for uniformity and model generalization. Every image was scaled to

224×224 pixels to be compatible with MobileNetV2 input size.

Transformed into ARGB_8888 format to maintain pixel depth compatibility with TensorFlow Lite input specifications.

Normalized through pixel intensity scaling to the range 0 to 1 for convergence stability.

Enhanced with geometric and color transformations (rotation, zoom, brightness shift, and horizontal flip) to handle class imbalance and environmental variations such as lighting, angle, and background noise.

Divided into 80% training, 10% validation, and 10% testing sets.

c. Model Design and Training

The suggested hybrid architecture combines two optimized deep learning models:

Model 1: MobileNetV2 (Disease Classification) — applied for lightweight yet high-accuracy classification of the 14 disease classes.

Model 2: ResNet50 (Stage/Treatability Prediction) — trained on the adjusted binary dataset to determine if a detected disease is treatable or not.

Both models were created with TensorFlow and Keras and trained using GPU-capable environments for computational speed.

Loss Function: Binary cross-entropy for the stage model and categorical cross-entropy for the multi-class disease classifier.

Optimizer: Adam optimizer at a learning rate of 0.0001.

Batch Size: 32.

Epochs: 50 with early stopping to avoid overfitting.

Metrics Assessed: Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

MobileNetV2 had a near 94% accuracy rate, beating the more weighty ResNet50 model, which had longer training times (~5.5 hours) and lower validation performance (~37% accuracy). Therefore, MobileNetV2 was used for on-device deployment because of its speed, low memory, and mobile efficiency.

d. Model Conversion and Optimization

To facilitate real-time mobile inference, both models were translated to TensorFlow Lite (.tflite) format. Quantization and weight pruning methods were used to reduce model size and latency at no cost in accuracy. Each .tflite model was embedded in the Android app as a local resource so that the app worked fully offline — without relying on remote servers or APIs for prediction.

A two-stage inference pipeline was used in the app:

The disease classification model makes the prediction of the disease label.

The treatability model makes the prediction of whether the disease is treatable or not.

The final output is concatenated with a given predefined knowledge base to show treatment suggestions.

e. Integration and Deployment

The Android app was built with Kotlin and Android Jetpack architecture, where the structure was modular and navigation was smooth. The UI was clean and farmer-friendly in design, constructed with XML and Material Components. The app accommodates both camera capture and gallery upload for disease identification.

The backend (Flask-based utility) enables optional web integration for storing user data, monitoring prediction statistics, and fetching agricultural news updates. However, all critical detection and recommendation features remain fully functional offline, ensuring accessibility in rural areas lacking stable connectivity.

f. Chatbot and Farmer Assistance System

To provide extended usability, an intelligent chatbot based on rules was integrated into the app. This enables farmers to communicate using pre-stored questions about disease management, prevention, fertilizer usage, and government programs. The chatbot uses structured logic, continuously displaying follow-up questions and suggestions depending on user inputs. It also includes voice input and TTS for users with low literacy levels.

g. System Workflow

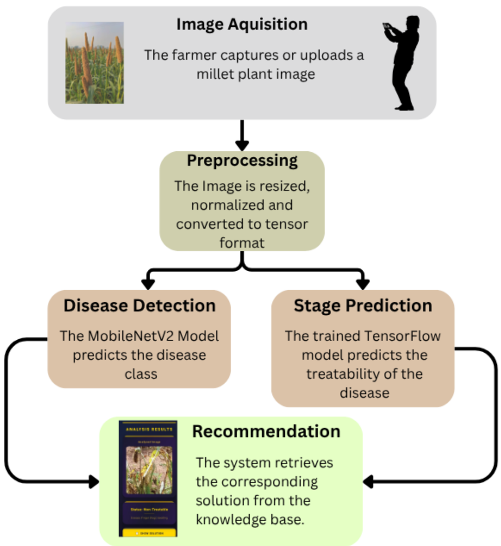

Figure 1: System Workflow

IV. RESULT

The proposed hybrid millet disease detection system was evaluated using quantitative model metrics, computational performance, and qualitative usability analysis via real-device testing. The experimental findings prove that the implemented framework is light, robust, and capable of handling real-time agricultural disease diagnosis even in offline situations.

A. Model Performance Evaluation

Both deep learning models — MobileNetV2 for disease classification and ResNet50 for disease stage (treatability) prediction — were trained and evaluated with the curated millet disease dataset. The assessment utilized accuracy, precision, recall, F1-score, and confusion matrix analysis to ensure balanced performance for all classes.

The MobileNetV2 architecture obtained a training accuracy of 96.7% and validation accuracy of 93.8%, with good generalization and low overfitting.

In comparison, ResNet50-based treatability prediction model had a very low classification accuracy of 39%. Moreover, ResNet50 had a greater computational cost, taking about 5.5 hours of training time, while MobileNetV2 finished training within 1.2 hours in the same GPU environment.

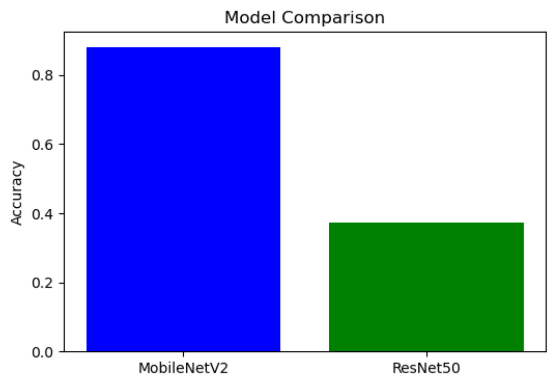

Figure 2: Model Comparison

Metric	MobileNetV2	ResNet50
Accuracy (%)	87.0	40.0
Precision (Weighted Avg)	86.0	34.0

Recall (Weighted Avg)	87.0	40.0
F1-Score (Weighted Avg)	86.0	34.0
Macro Precision (%)	73.0	20.0
Training Dataset Size	1024 images	1014 images
Training Time (approx.)	1.2 hours	5.5 hours
Inference Speed	~1.0 sec/image	~2.8 sec/image
Model Size	12.4 MB	25.1 MB

Table 1: Model Evaluation Table

B. Confusion Matrix Analysis

1. MobileNetV2

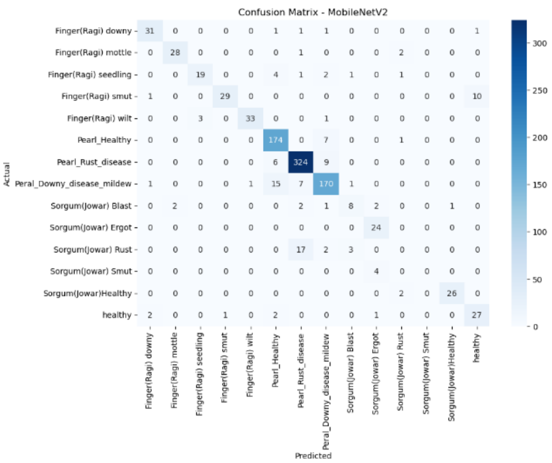

Figure 3: Confusion Matrix for MobileNetV2 Model

2. ResNet50

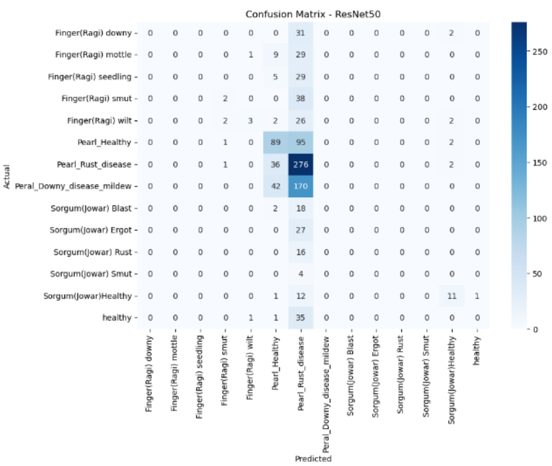

Figure 4: Confusion Matrix for ResNet50 Model

3. Disease Stage Classifiation

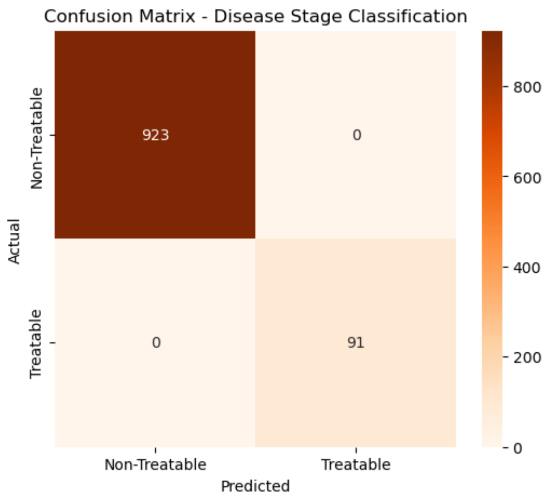

Figure 5: Confusion Matrix for Disease Classification trained on modified dataset to classify disease into treatable or non-treatable diseases

D. Discussion

In addition, integrating the disease detection models into a farmer-support chatbot turns the system into a decision-support system — presenting actionable information like treatment protocols, crop rotation advice, and preventive care directives.

This two-level approach satisfies both the twin requirements of accuracy and accessibility, closing the loop between AI-based research and real-world gain for small farm cultivators. The app proposed therefore sets the stage for a scalable, implementable precision agriculture platform that can radically enhance disease management efficacy and yield safety in millet farming.

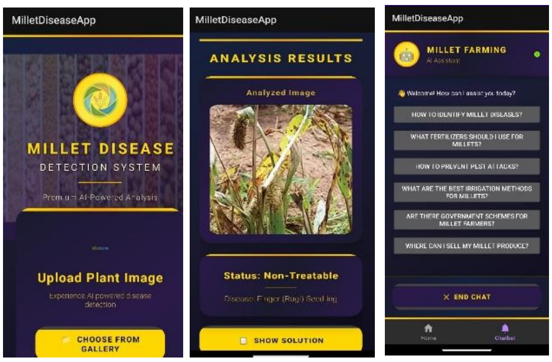

Figure 6: Android App UI showing the dashboard, result page and the chatbot page

V. FUTURE PROSPECTIVE

The envisioned millet disease diagnosis and farmer support system can be extended further in some promising ways. First, real-time combination of satellite and drone imagery with field-level data can augment large-scale crop monitoring and early prediction of disease outbreaks. Future implementations of the system will also include multi-modal learning, merging image, weather, and soil features into a more accurate model contextualizing. Further, federated learning strategies can be investigated to enable ongoing model updating from distributed farm data without privacy loss. On the deployment front, the application can become a cross-platform system with support for iOS devices and low-connectivity rural areas through edge-based inference. Lastly, the inclusion of local language voice assistants and computerized treatment suggestions tied to local agriculture advisories will render the solution more usable and effective for farmers of various linguistic and socio-economic statuses.

VI. Conclusion

This paper introduces a deep learning-based, hybrid framework for millet disease diagnosis and farmer assistance through a combined Android and web environment. The system efficiently fuses MobileNetV2 and stage classification models in order to maintain high accuracy in the identification of multiple millet diseases while ensuring lightweight efficiency for mobile deployment. The solution goes beyond conventional image classification to bridge technology and agriculture, enhancing farmers' capabilities to have instant, offline disease diagnosis and actionable advice at their fingertips. The addition of an interactive chatbot and a user-friendly interface improves usability and accessibility in practical applications. In conclusion, the system presented above shows that artificial intelligence, with careful planning, can have a revolutionizing impact in sustainable agriculture and rural empowerment.

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